MT-C Series Wi-Fi 7 Client Emulators

MT-C4 · MT-C12 · MT-C24 · MT-C32

Overview

The MT-C Series is a family of Wi-Fi 7 client emulators designed for testing and validating next-generation 802.11be devices. Every model operates across all three Wi-Fi bands (2.4, 5 and 6 GHz), supports channel bandwidths up to 320 MHz, and provides full Wi-Fi 7 capabilities including Multi-Link Operation (MLO) and Extremely High Throughput (EHT), with backward compatibility down to Wi-Fi 5/6/6E.

The series scales from 4 individual radios (MT-C4) to 32 individual radios (MT-C32), allowing the same software, automation and analysis environment to be used from bench-top development through to high-density performance and stress testing. The MT-C instruments support Intel, MediaTek and Qualcomm based Wi-Fi 7 stations, enabling true multi-vendor client ecosystems against the access point under test.

Beyond iPerf-based TCP, UDP and bi-directional traffic, MT-C emulators run real application traffic — Microsoft Teams, Google Meet, Zoom and YouTube — on the emulated clients, and collect live per-application quality-of-experience statistics alongside per-station Wi-Fi link metrics. Every test concludes with a self-contained interactive report generated by the MT-CORE automation suite.



MT-C24 — 24-radio unit with front-panel SMA RF ports (RF1–RF24)



MT-C4 — compact 4-radio bench-top unit

Key Features

- **4, 12, 24 or 32 individual Wi-Fi 7 radios** from a single chassis
- **Intel, MediaTek and Qualcomm based Wi-Fi 7 stations** — emulate mixed, real-world client ecosystems
- **Wi-Fi 7 (802.11be) support** with full backward compatibility
- **Multi-Link Operation** — MLMR and eMLSR modes
- **Traffic generation**: iPerf TCP, UDP and bi-directional
- **Real application traffic emulation**: Microsoft Teams, Google Meet, Zoom and YouTube with per-application statistics
- **Completely shielded** from outside interference
- **Modular architecture**, easily scalable to more radios
- **Advanced analysis** and traffic generation tools
- **Completely automated** with interactive HTML test reports
- **Sniffing** along with real-time analysis

Model Comparison

Parameter	MT-C4	MT-C12	MT-C24	MT-C32
Individual radios	4	12	24	32
Max. emulated Wi-Fi 7 stations	4	12	24	32
Station chipset options	Intel / MediaTek / Qualcomm	Intel / MediaTek / Qualcomm	Intel / MediaTek / Qualcomm	Intel / MediaTek / Qualcomm
Frequency bands	2.4 / 5 / 6 GHz	2.4 / 5 / 6 GHz	2.4 / 5 / 6 GHz	2.4 / 5 / 6 GHz
Max. channel bandwidth	320 MHz	320 MHz	320 MHz	320 MHz
MLO modes	MLMR, eMLSR	MLMR, eMLSR	MLMR, eMLSR	MLMR, eMLSR
Emulation modes	Client, Sniffer, Mixed	Client, Sniffer, Mixed	Client, Sniffer, Mixed	Client, Sniffer, Mixed
Typical application	Bench-top development and functional testing	Feature and regression validation	Performance, roaming and mesh test systems	High-density capacity and stress testing

All models share the same MT-CORE software, automation interfaces, analysis and reporting environment.

Common Specifications

Parameter	Value
Radios	4 / 12 / 24 / 32 individual radios (model dependent)
Station chipsets	Intel, MediaTek and Qualcomm based Wi-Fi 7 stations
Bands	2.4 GHz, 5 GHz and 6 GHz
Operating frequency	2400–2485 MHz 5150–5250 MHz 5250–5350 MHz 5470–5725 MHz 5725–5895 MHz 5925–7125 MHz
RF output power	20 dBm (2400–2485 MHz) 23 dBm (5150–5725 MHz) 21 dBm (5725–5875 MHz, 5925–7125 MHz)
RX sensitivity	2.4 GHz: −97 dBm · 5 GHz: −95 dBm · 6 GHz: −95 dBm
Emulation modes	Client, sniffing and mixed client emulation
Bandwidth	20, 40, 80, 160, 320 MHz and sub-channel RUs
Standards	802.11be (Wi-Fi 7), backward compatible with Wi-Fi 6/6E/5
MLO modes	MLMR, eMLSR
Monitoring	RSSI, MCS, Nss, bandwidth plus per-packet deep performance metric plots
Sniffer	Wireshark captures
802.11ax PHY	DL/UL OFDMA
802.11ax MAC	Trigger frame support Non-trigger-based and trigger-based sounding for beamforming

Parameter	Value
	UL-OFDMA random access
802.11be PHY	DL/UL OFDMA, DL/UL MU-MIMO
TX power	MCS, streams, frequency and channel-width dependent
Encryption	Open, WPA, WPA2 and WPA3
Traffic generation	iPerf: TCP, UDP and bi-directional Real application traffic: Microsoft Teams, Google Meet, Zoom, YouTube — with live per-application statistics collection
Reporting	Automated, self-contained interactive HTML reports (MT-CORE)

Configurable Parameters

- IEEE 802.11 standards
- Band and channel
- Multi-Link Operation
- Frame aggregation on/off
- Number of streams
- Power level — high, mid and low
- Roaming threshold
- 802.11 k/v/r
- Different security types
- Guard interval
- Power save and packet length

Traffic Generation and Real Traffic Emulation

Each emulated station can generate synthetic traffic or run genuine application sessions, so an access point can be validated not only for raw throughput but for the quality of experience it delivers to real applications under load, interference and mobility.

iPerf Traffic

- **TCP** — uplink and downlink throughput testing per station
- **UDP** — configurable rate, packet length and streams
- **Bi-directional** — simultaneous uplink and downlink traffic on the same station

Real Application Traffic

Emulated clients run live sessions of the applications below while the instrument collects per-application quality metrics in real time, alongside the underlying Wi-Fi link statistics:

Application	Statistics collected
Microsoft Teams	Video: bitrate, frame rate, RTT, received packets Audio: bitrate, RTT, jitter, packet loss, received packets, MOS Screen share: bitrate, frame rate, RTT, packets, packet loss, processing time
Google Meet	Video, audio and screen-share session quality statistics
Zoom	Audio: frequency, jitter, latency, packet loss, MOS Video: frame rate, jitter, packet loss Screen share: frame rate, latency, jitter, packet loss

Application	Statistics collected
YouTube	Playback FPS, video bitrate, cache buffer duration

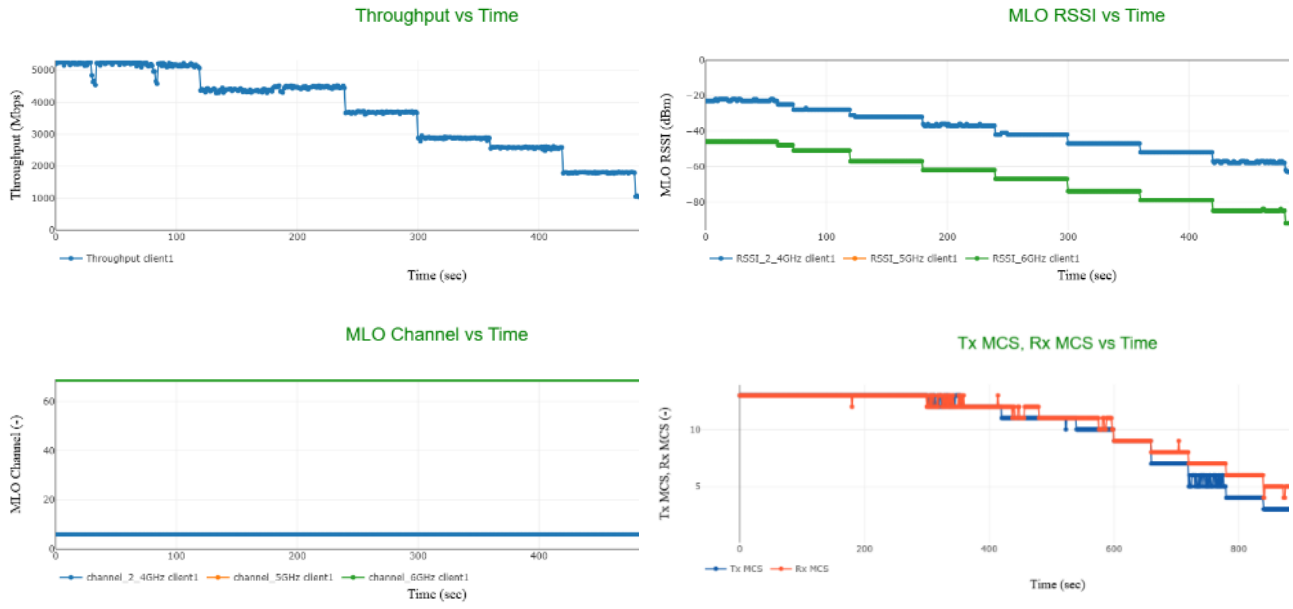
Statistics Collection and Monitoring

All metrics are recorded per station and per MLO link versus time, plotted live during execution and preserved in the test report for post-analysis:

Category	Metrics
Throughput and traffic	Throughput, TX/RX data rate, TX/RX packets, RX drops, TX retries, TX failures, ping latency
RF and PHY	RSSI, operating band, TX/RX MCS, TX/RX NSS, TX/RX bandwidth, guard interval, preamble
BSS parameters	Beacon interval, DTIM period
MLO	Per-link RSSI, channel, band, data rate and MCS for each active link
Application QoE	Teams / Meet / Zoom audio, video and screen-share quality metrics; YouTube playback statistics

Real-Time Analysis

Per-station and per-link metrics are plotted live during test execution — throughput, TX/RX MCS, MLO RSSI and data rate per link — with per-packet deep performance metrics available for post-analysis.



Interactive Test Reports

Every test run automatically produces a self-contained interactive HTML report through the MT-CORE automation suite — a single file that can be archived, e-mailed and opened in any browser with no server or installation required.

- **Test information and configuration** — test type, DUT model and firmware version, traffic type, band, duration and attenuation profile are captured automatically
- **Interactive plots** — zoom, pan and hover on every chart; enable or disable individual station traces
- **One-click metric filters** — filter the measurement set by RSSI, band, bandwidth, MCS, NSS, MLO and more

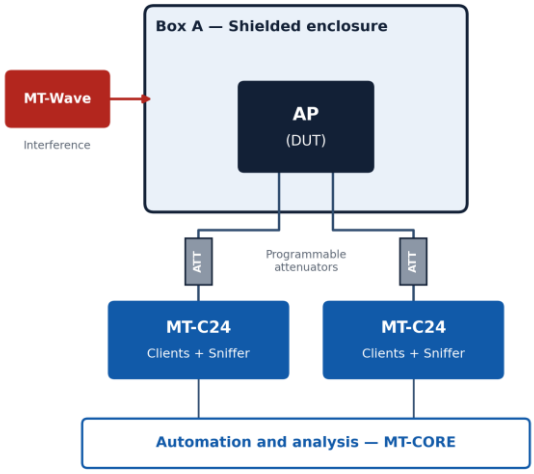
- **Complete statistics record** — all Wi-Fi link, MLO and application QoE statistics plotted versus time for every station
- **Built-in automated test suites** — e.g. Rate vs Range with programmable attenuation steps between AP and stations

Applications

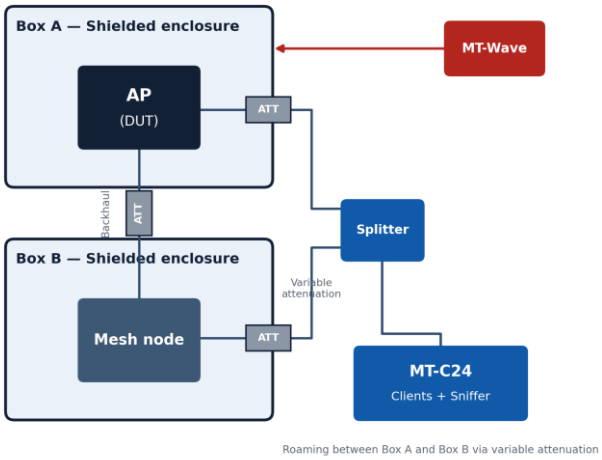
- **Wi-Fi 7 AP Testing:** Perform comprehensive validation of Wi-Fi 7 access points under real-world scenarios with Intel, MediaTek and Qualcomm client mixes.
- **Multi-Link Operation (MLO) Validation:** Ensure seamless MLO across multiple bands, optimising bandwidth and latency.
- **Throughput and Latency Testing:** Evaluate device performance in terms of data rate, response times and overall network efficiency.
- **Application QoE Testing:** Measure real video-conferencing and streaming quality — Teams, Meet, Zoom and YouTube — delivered through the AP under test.
- **Rate vs Range Testing:** Characterise baseband and RF chain performance versus range using programmable attenuation.
- **Wi-Fi Coexistence Testing:** Test how devices handle interference from other Wi-Fi networks and legacy technologies.
- **Stress Testing:** Simulate network congestion and client saturation to observe performance under extreme conditions.

System Configurations

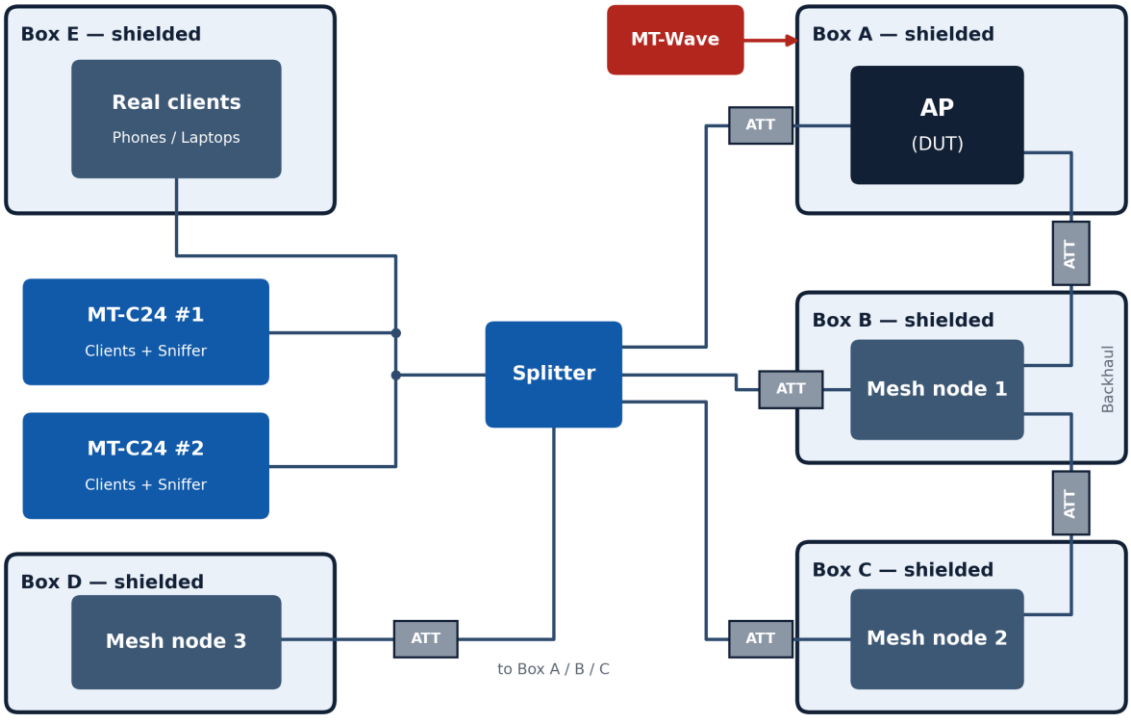
MT-C Series emulators integrate with Wavemetrik shielded enclosures, programmable attenuators, splitters and the MT-Wave interference generator to build fully conducted, repeatable test systems — from a single-enclosure performance setup to multi-enclosure roaming, mesh and 48-radio performance systems.



Single-enclosure performance test system



Mesh and roaming test system



Performance and mesh test system — 48 radios with real clients, interference and multi-hop backhaul



Integrated multi-enclosure test system with MT-C Series instrumentation